DSA REFERENCE

# Monotonic Stack

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Keep unresolved candidates in monotonic order; a new value resolves every dominated candidate once.

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DSA / MONOTONIC-STACK

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# Monotonic Stack

**CORE IDEA**

**Keep unresolved candidates in monotonic order; a new value resolves every dominated candidate once.**

### Recognition cues

+ next greater or smaller
+ span
+ histogram area
+ nearest boundary

### Implementation sequence

+ store indices
+ choose strict or non-strict comparison
+ resolve during pops
+ flush with a sentinel when necessary

## Invariant

Stack indices remain monotonic by value and each stored index is still waiting for its resolving boundary.

## Worked reasoning

For next greater element, pop smaller values when a larger value arrives and assign the current index as their answer.

## Complexity

O(n) time because each index is pushed and popped at most once; O(n) space.

## Failure modes

- [ ] wrong equality rule
- [ ] storing values without positions
- [ ] missing final flush
- [ ] confusing previous and next boundaries

Practice ladder

# Monotonic Stack: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

| # | Problem | Deliberate-practice prompt |
|---|---|---|
| 1 | **Daily Temperatures** | State the cue, invariant, edge case, and proof before implementation. |
| 2 | **Next Greater Element I** | State the cue, invariant, edge case, and proof before implementation. |
| 3 | **Largest Rectangle in Histogram** | State the cue, invariant, edge case, and proof before implementation. |
| 4 | **Car Fleet** | State the cue, invariant, edge case, and proof before implementation. |
| 5 | **Sum of Subarray Minimums** | State the cue, invariant, edge case, and proof before implementation. |

## Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

## Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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